REXUS BRYAN L. GAN

CAD & DESIGN ENGINEERING

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PROFESSIONAL SUMMARY

Aeronautical Engineering graduate (BS, 2026) specializing in parametric CAD, with FreeCAD as my primary tool and working knowledge of SolidWorks. Portfolio includes a complete trainer-aircraft design (GANNET-T731), a Python-based aircraft design tool, and a rim-driven propeller study. Experienced in explaining technical content as Education and Research Director for two student organizations. Seeking to teach practical CAD workflows.

TECHNICAL SKILLS

CAD & Drafting: FreeCAD (parametric modeling, TechDraw), SolidWorks (basic–intermediate modeling and drawings), OpenVSP; 3D part and assembly modeling, technical drawing

Aeronautical Engineering: Conceptual aircraft/UAV design, requirement-based sizing, weight estimation, airfoil selection, propeller and climb-performance analysis, wing optimization

Simulation & Analysis: OpenFOAM, SALOME, SimScale, XFLR5; external aerodynamics and CFD fundamentals

Programming & Tools: Python (engineering calculations, automation), Git/GitHub, Blender (animation and rendering), AI-assisted development.

PROJECTS

Conceptual UAV

- Developed the conceptual design, covering configuration selection, component placement, and preliminary weight estimation
- Solved the coupled mission-weight equation in Python using fixed-point iteration, since the weight appears on both sides of the equation
- Modeled the geometry in OpenVSP, produced technical drawings in FreeCAD TechDraw, and rendered presentation visuals in Blender

Aircraft Design (GANNET-T731)

- Sized the aircraft against stall, maximum speed, take-off, climb, and ceiling requirements using Python (Matplotlib) constraint plots, giving a maximum wing loading of 14.89 lbf/ft^2 and power loading of 8.28 lbf/hp
- Developed a weighted airfoil-selection scheme with separate criteria for the root, middle, and tip sections and ran multiple configurations in xfoil automated with python
- Built a parametric OpenVSP model and ran low-fidelity aerodynamic analysis using VSP Aero
- Optimized wing geometry (taper, twist, span distribution) using Response Surface Methodology
- Coupled propeller-efficiency, excess-power, and climb-performance equations, and automated the full workflow in Python

Rim Driven Toroidal Propeller

- Learned and applied basic to intermediate parametric part modeling and technical drawing in SolidWorks, in my first project with the software
- Simulated CV and TV configurations in SimScale, comparing torque, thrust, and advance ratio
- Limited the study to low advance ratios to stay within the platform's mesh-count, run-time, and fidelity limits.

Technical Drawing Series

- Produced 13 dimensioned drawing sheets (9 in FreeCAD, 4 in SolidWorks) with orthographic, section, and isometric views of aircraft, a landing-gear mechanism, and machine components

Fulcrum: Board Exam Review Platform (AI-assisted)

- Developed a browser-based review platform for engineering board exam preparation, with a question bank and interactive learning modes
- Built with AI-assisted development and published on GitHub Pages

FlightForge: Python Aircraft Design Tool (AI-assisted)

- Built the PyQt application framework, then developed the GUI with AI assistance
- Integrated parametric wing and fuselage modeling, OpenFOAM CFD, V-n diagrams, and XML export for FlightGear
- Wrote the engineering scripts for weight sizing, power and wing-loading plots, and weighted airfoil selection

Blender Animation

- Animated CAD-exported models in Blender using keyframes, including a rim-driven propeller concept video
- Automated importing images into a Blender scene with a Python script (AI-assisted)

EDUCATION

Mats College of Technology | Bachelor of Science in Aeronautical Engineering
Graduation year: 2026

AWARDS & RECOGNITION

- Best in Aircraft Design — Department of Aeronautical Engineering, 2026 (GANNET-T731)
- Best in Feasibility Study— Department of Aeronautical Engineering, 2026 (Flight Forge)
- Best in UAV Design — Department of Aeronautical Engineering, 2026 (KingFisher-R31)

AFFILIATIONS

Mindanao Aeronautical Engineering Student Organization (MAESO), 2021-2026

- Education and Research Director (2024-2026): facilitated and tutored weekly sessions on core Aeronautical Engineering subjects
- 2nd Year Representative (2022-2023) and 1st Year Representative (2021-2022): took part in organization activities

Society of Aerospace Engineering of the Philippines (SAEP SCF), 2021-2022 | Education and Research Director

- Ran *dAELY Trivia*, a daily educational Facebook series that shared aeronautical engineering trivia with students

Society of Engineering and Technology, 2025-2026 | Info Media

REFERENCE

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